RESEARCH INTRODUCTION · FALL 2027 PHD

From Embodied Understanding to Reliable Robotic Action

Research experience / Future directions

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RELIABLE EMBODIED AI

OVERVIEW

My Research Experiences

01

ENACT

World-state understanding

Contributor · ICLR 2026

Action → world change

02

MindTopo

Topological reasoning

Contributor · NeurIPS 2026 workshops

Recognition → planning

03

TopoGym → RoboARC

Physical execution

Co-first author (equal contribution)

Reasoning → manipulation

RoboARC: CVPR 2027 submission target, work in progress

ENACT: Can foundation models anticipate action consequences?

Egocentric world modeling: predict how actions change an environment.

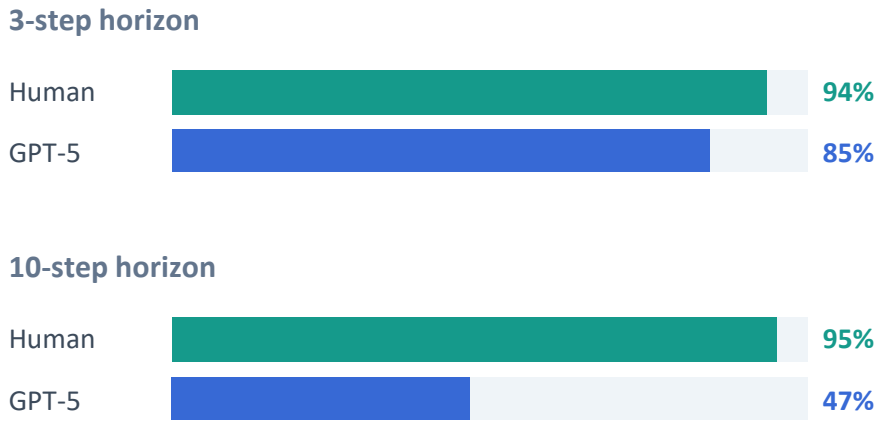
ACTION-CONDITIONED WORLD UNDERSTANDING



Forward: actions → observations

Inverse: observations → actions

Forward pairwise accuracy

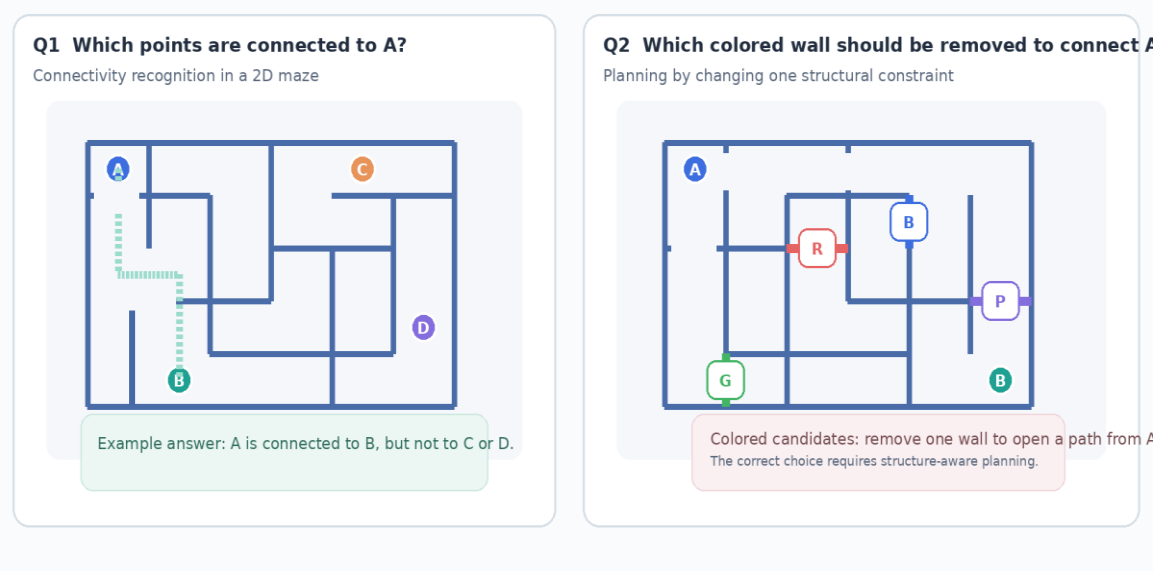


Source: ENACT (ICLR 2026) • rounded display values

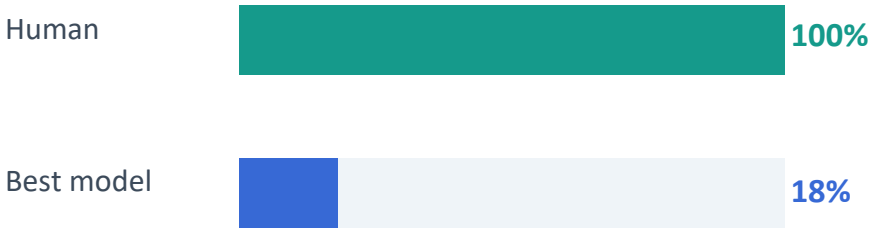
MY CONTRIBUTION Sim-2-Real Data collection + Part of Ablations + Failure analysis

MindTopo: Recognizing topology and acting

What remains invariant?



Easy “Untangle” — task success



13 / 13 models plan worse than they reason

Source: mind-topo.github.io (Easy task subset)

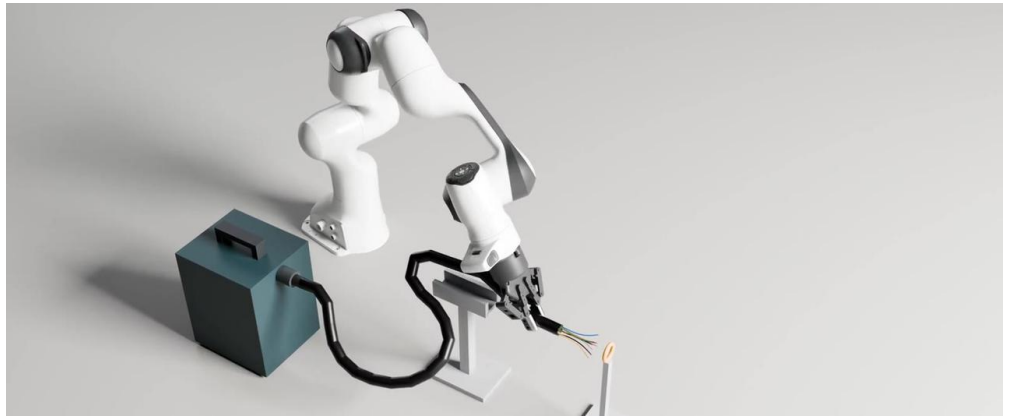
MY CONTRIBUTION Built 2D maze task + Establishing Evaluation servers + Part of Human annotation

From topological puzzles to executable policies



TopoGym (a previous prototype) prototype)

Do robotic policies turn topological reasoning into physical manipulation?

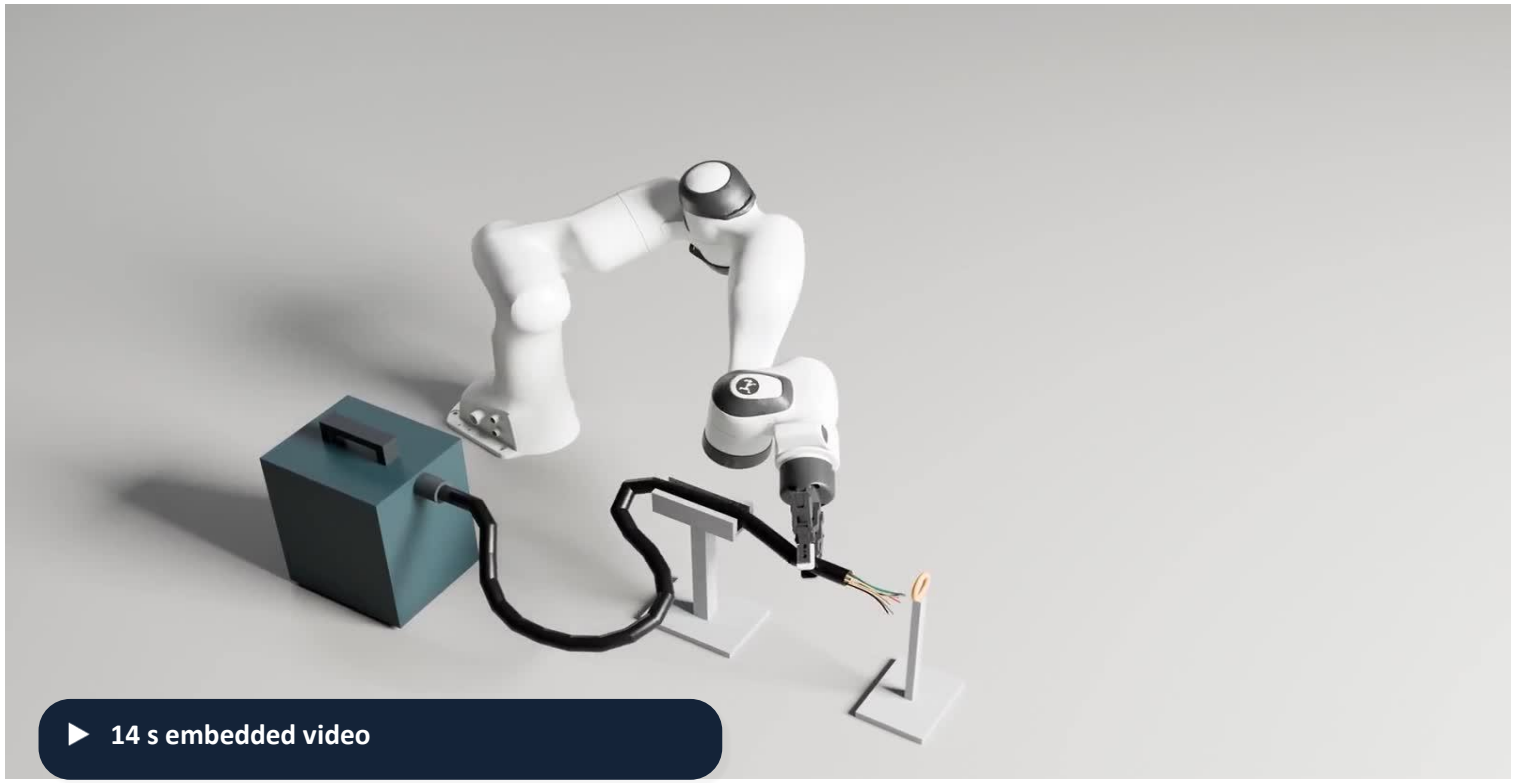


RoboARC

Can coding agents generate executable policies under continuous constraints?

SCOPE Interlocking · Narrow passages · Threading · Assembly · Deformables

Physical constraints persist across thousands of steps



90 task trails (per model)

Task Families



Information setup modes

Privileged: all info & rewind

Standard: only visual info

Source (RoboARC in EFE): [mll-lab-nu/embodied-first-exam: The first exam for code agents in the physical world: frozen robot tasks from many benchmark suites, each answered with a program and verified in a separate sandbox. Runs on Harbor.]

The privileged-to-standard gap is large

Early publicly reported subset; full benchmark evaluation is ongoing.

E.g. GPT-6 Astra · earlier 58-task snapshot



~78% vs ~19%

Not a perception-only ablation:
mode also changes resets & access.

MY CONTRIBUTIONS

01 Idea formulation & task design

02 Simulator and task implementation

03 Reference / oracle solutions

04 Engine migration & evaluation follow-up

NEXT: All 90 bases × 2 modes × 3 trials runs for each model + Analyze environment variants & reasoning–execution gap

Research Plan: Reliable perception + adaptive manipulation

OBSERVE

Sensors + context

→

REASON

Risk / physics beliefs

→

ACT

Probe or intervene

→

ADAPT

Update the policy

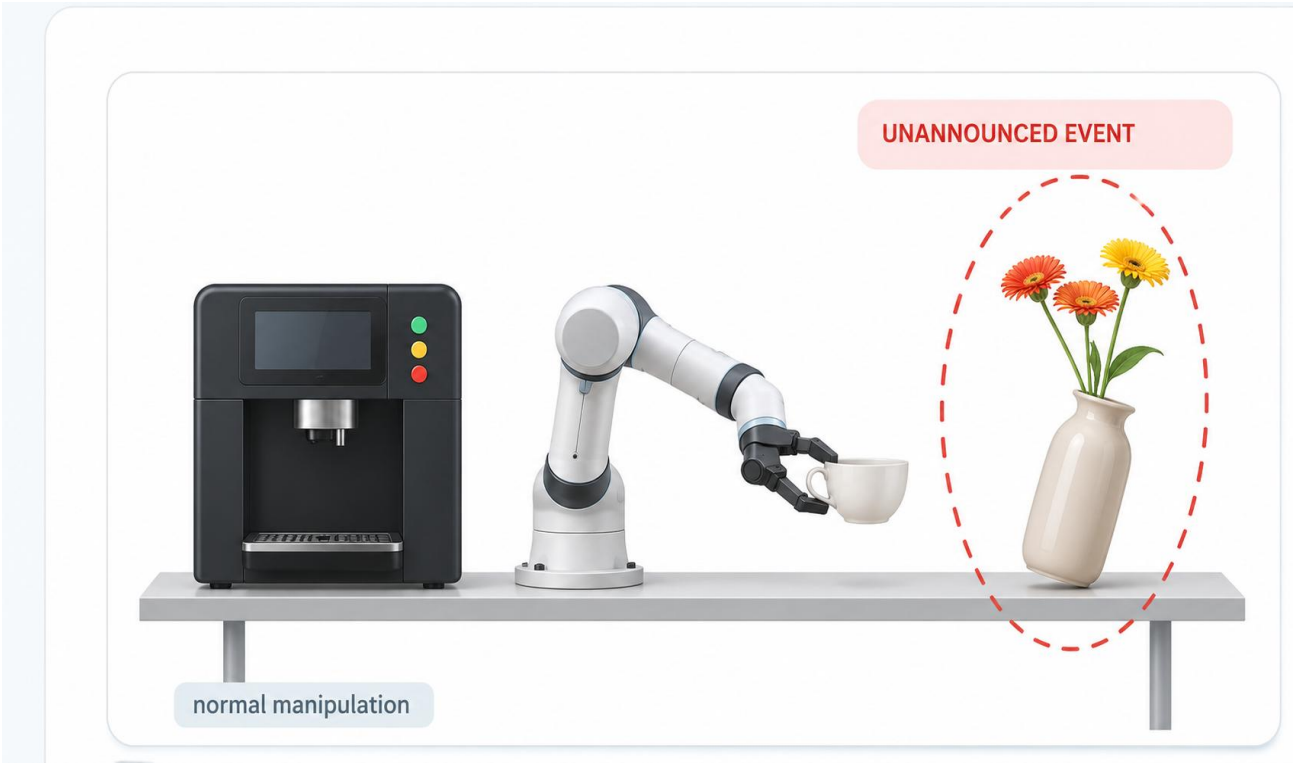
DIRECTION 01

Unannounced hazards

DIRECTION 02

Online physical modeling

Will a robot notice an unrelated & unannounced danger?



Possible response loop

01 **Detect**
Motion / instability

02 **Intervene**
Safe closed-loop action

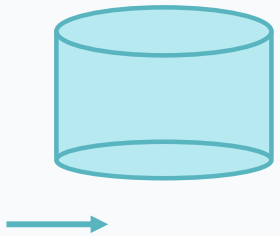
03 **Resume**
Return to main task

Benchmarking: **unannounced**, **task-irrelevant** risk awareness + intervention and safe recovery to task

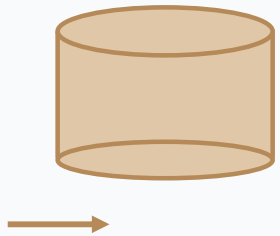
Learn about the manipulated system during rollout (with World-Model)

Start with physical priors → test uncertain hypotheses → adapt the policy using contact feedback.

Same geometry with different hypotheses



smooth-looking



rough-looking

Appearance gives priors—not exact physical properties. properties.

01 Infer — geometry, mass, friction, contact (from World-Model) + current location & gesture

↓

02 (may) Probe — small, safe physical tests

↓

03 Update — belief / world model online

↓

04 Adapt — planning or VLA action guidance

Proposed shift: object/contact dynamics → manipulation

Selected papers & project resources

Related work selected for relevance and methodological transfer; citations link to source papers.

ENACT

- Wang, Q. et al. (2026). ENACT: Embodied Cognition with Egocentric World Modeling. ICLR.

MINDTOPO

- Ge, Y. et al. (2026). MindTopo: Can Foundation Models Reason in Topological Space? NeurIPS Workshops.

TOPOGYM → ROBOARC

- RoboARC (2026). [paper in preparation for CVPR 2027, arxiv planned on late Nov].

DIRECTION 01 · UNANNOUNCED HAZARDS

- Li, Y. et al. (2026). ReactHuman: Physics-Grounded Reactive Decision-Making. arXiv:2609.10895.
- Luo, J. et al. (2026). SafeStage: Safety Before, During & After Manipulation. arXiv:2609.21223.
- Zhu, X. et al. (2026). ProAct: Structure-Aware Proactive Response. ICML.
- Xu, T. R. et al. (2026). RT-Safe: Real-Time Embodied Agent Safety. arXiv:2610.09294.

DIRECTION 02 · ONLINE WORLD MODELING

- Çiçek, B. et al. (2026). CoRAL: Contact-Rich Adaptive LLM Control. arXiv:2605.02600.
- Li, W. et al. (2025). PIN-WM: Physics-Informed World Models for Non-Prehensile Manipulation. RSS.
- Liao, Q. et al. (2026). SuperMimic: Humanoid Control with One Differentiable World Model. CoRL.